

SAR Lab 7

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1 Steps taken

- Downloaded pre-event COMET-LiCS interferograms (descending frame 062D_05831_131313, cutoff 20170624, max baseline 72 days).
- Subset around the landslide and converted to a MintPy stack (AOI 103.55–103.65°E, 31.55–31.65°N).
- Ran SBAS (`smallbaselineApp.py`) to get the velocity field and displacement time series (auto reference point).
- Mapped the landslide with co-event coherence (LiCS pair 20170607_20170725) and with Sentinel-1 VV change in GEE.
- Used Sentinel-2 (NDVI, BSI) to map the affected area and quantify the surface change.

2 SAR results

SBAS shows progressive pre-failure motion on the slope: a pixel at 31.576°N, 103.563°E moves at 1.28 ± 0.13 cm/yr over 2014–2017 (Fig. 1). The automatic reference point already produced a clean trend with no dominant seasonal signal, so it was retained.

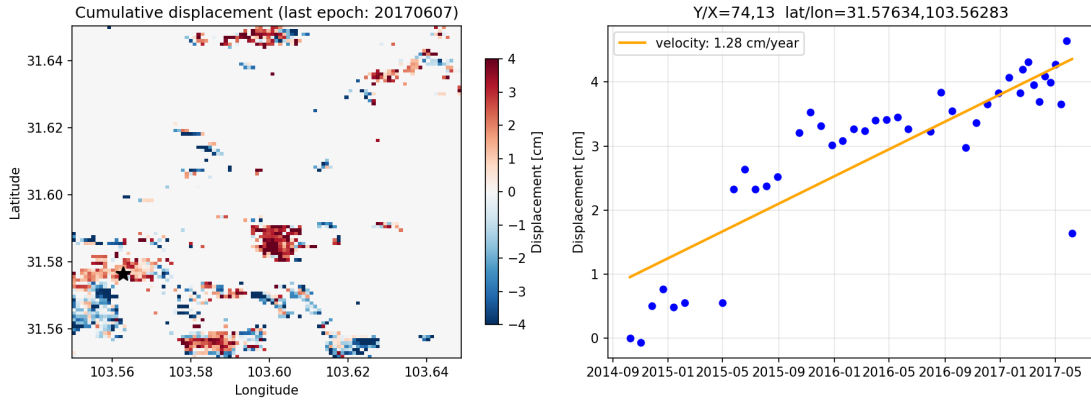


Figure 1: SBAS displacement map (left) and time series (right) for the failing slope; near-linear pre-failure creep of ~ 1.3 cm/yr.

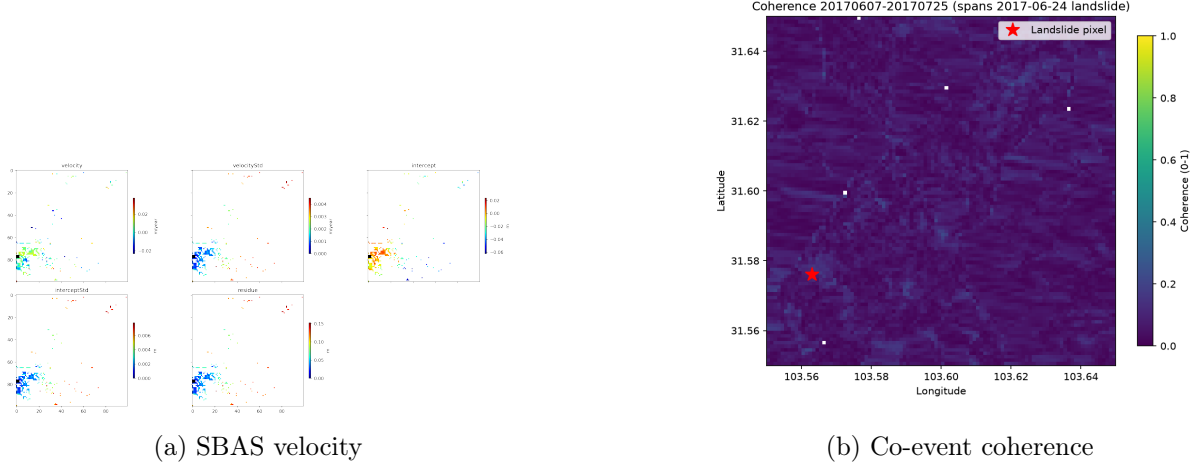


Figure 2: Left: mean LOS velocity. Right: coherence is uniformly low (mean ≈ 0.06) over the whole AOI.

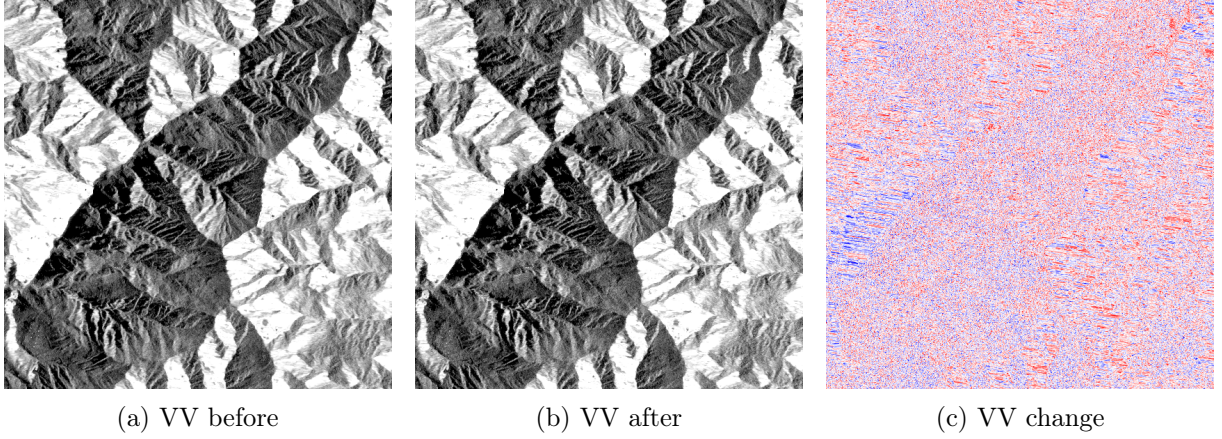


Figure 3: Sentinel-1 VV backscatter before/after the event and their difference.

Coherence vs. VV. Coherence is *not* useful here: dense vegetation, steep relief and the long baseline cause near-total decorrelation everywhere, so the landslide cannot be separated from the background. The VV amplitude change is *no better*: the difference image is dominated by speckle and layover in this steep terrain, and does not clearly isolate the failure either. In this case neither SAR change product reliably maps the landslide.

3 Optical results

The post-event Sentinel-2 image shows a fresh, bright scar that delineates the affected area (Fig. 4). NDVI drops strongly over the scar (vegetation loss) and BSI increases (newly exposed bare ground/rock).

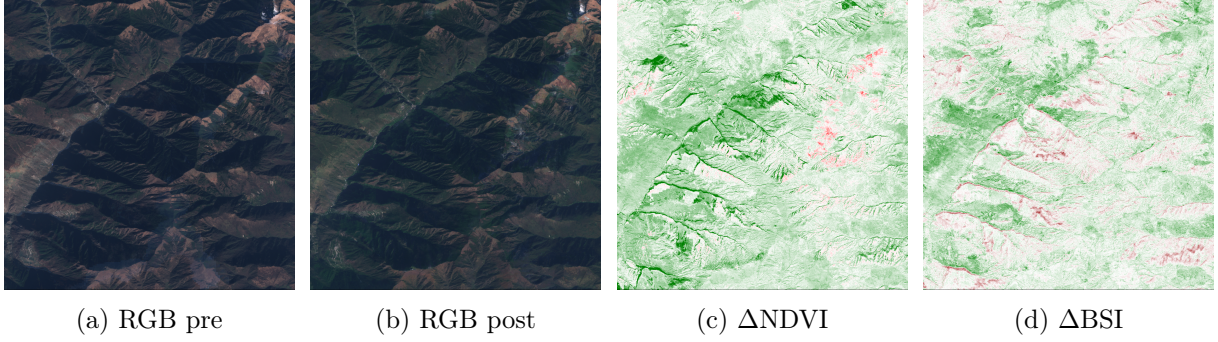


Figure 4: Sentinel-2: RGB before/after, NDVI change (red = vegetation loss) and BSI change (brown = exposed soil) over the landslide.

4 Conclusions

SBAS recovers the slow pre-failure creep (~ 1.3 cm/yr). For mapping the landslide itself, both SAR change products fail in this vegetated, steep terrain: coherence is uniformly low and the VV change is dominated by speckle. The Sentinel-2 optical indices (NDVI loss + BSI increase) clearly delineate and quantify the surface change and are by far the best mapping tool in this case.